KAZAKH BRITISH TECHNICAL UNIVERSITY

Project report

Information theory

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Comparison of tracking and relevant data fusion algorithms using entropy based metrics

1. Introduction

In this project, we study the problem of multiple object tracking in aerial imagery. The main goal is to compare different tracking algorithms and evaluate their performance under realistic conditions.

In real-world scenarios, object detectors are not perfect: they may miss objects, introduce noise, or generate false detections. Therefore, it is important to evaluate how tracking algorithms behave under such conditions.

In this work, we compare three algorithms: JPDA, GM-PHD, and PMBM-lite. Their performance is evaluated using classical metrics such as precision, recall, F1-score, and also using KL divergence.

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2. Dataset

We use the M3OT dataset, which is designed for multi-object tracking in aerial imagery. The dataset contains video sequences captured by UAVs at different altitudes.

It includes:

- RGB and infrared (IR) images
- more than 20,000 frames
- over 200,000 annotated bounding boxes

Each frame contains ground truth annotations describing object positions.

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3. Methodology

3.1 Pipeline

The overall pipeline of the system is as follows:

Ground Truth → Simulated Detections → Tracking Algorithms → Evaluation

First, ground truth data is loaded from the dataset. Then, noisy detections are generated to simulate realistic conditions. After that, tracking algorithms are applied. Finally, the results are evaluated.

3.2 Simulation of Detections

To simulate real-world conditions, we intentionally degrade the perfect ground truth data.

Three types of imperfections are introduced:

- **Noise:** object positions are slightly shifted
- **Missed detections:** some objects are randomly removed (miss_prob = 0.2)
- **False detections:** additional fake objects are added (false_per_frame = 2)

This step ensures that the tracking problem is realistic and challenging.

3.3 Tracking Algorithms

JPDA (Joint Probabilistic Data Association)
JPDA associates detections with existing tracks using probabilistic matching. It considers multiple possible assignments between detections and objects.
However, JPDA tends to include false detections, which leads to a high number of false positives.

GM-PHD (Gaussian Mixture Probability Hypothesis Density)
GM-PHD does not explicitly track individual objects. Instead, it estimates the spatial density of objects using Gaussian mixtures.
This approach reduces false detections but often misses real objects, resulting in lower recall.

PMBM-lite (Poisson Multi-Bernoulli Mixture)
PMBM combines tracking and probabilistic modeling. It distinguishes between existing objects and potential new ones.
The "lite" version simplifies the full PMBM algorithm by reducing computational complexity while preserving its main idea.
PMBM provides a balance between detecting objects and avoiding false detections.

3.4 Kalman Filter
All algorithms use a Kalman filter to model object motion.
The Kalman filter predicts the position of objects in the next frame and updates this prediction using noisy observations. This allows the system to maintain tracking even when detections are missing.

3.5 Evaluation Metrics
The performance of the algorithms is evaluated using the following metrics:

- **Precision** – proportion of correct detections among all detections
- **Recall** – proportion of detected objects among all real objects
- **F1-score** – balance between precision and recall
- **Mean distance** – average localization error
- **KL divergence** – difference between predicted and ground truth distributions
- **TP (True Positives)** - correctly detected objects
- **FP (False Positives)** - incorrect detections (false objects)
- **FN (False Negatives) -** missed real objects

KL divergence is particularly important because it measures how well the algorithm models the overall distribution of objects.

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4. Results
The results for video_id = 9 are presented below:

Algorithm	**Precision**	**Recall**	**F1**	**TP**	**FP**	**FN**	**KL**
JPDA	0.127	0.998	0.226	1298	8911	3	3.23
GM-PHD	1.000	0.451	0.622	587	0	714	6.21

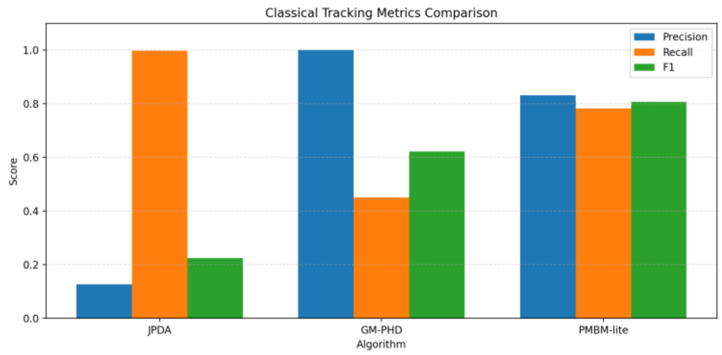
Algorithm	Precision	Recall	F1	TP	FP	FN	KL
PMBM-lite	0.831	0.783	0.806	1019	207	282	1.75

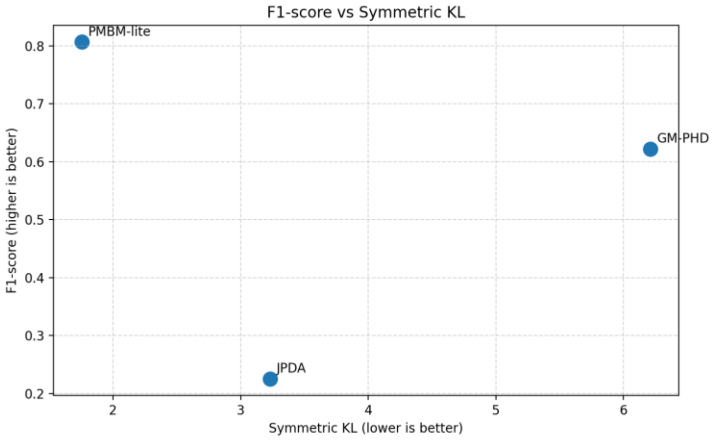
Lower KL divergence indicates that the predicted spatial distribution is closer to the ground truth.
Higher F1-score reflects a better balance between precision and recall.

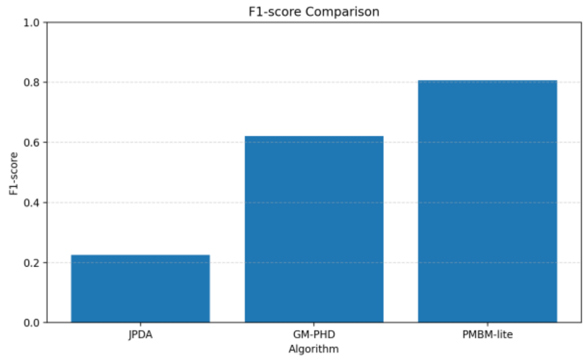
Observations

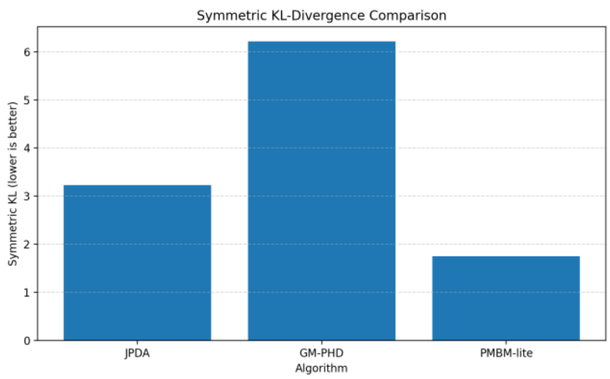
- JPDA achieves very high recall but produces many false positives
- GM-PHD has perfect precision but misses many objects
- PMBM-lite provides a good balance between precision and recall

PMBM-lite also achieves the lowest KL divergence, indicating the closest match to the ground truth distribution.









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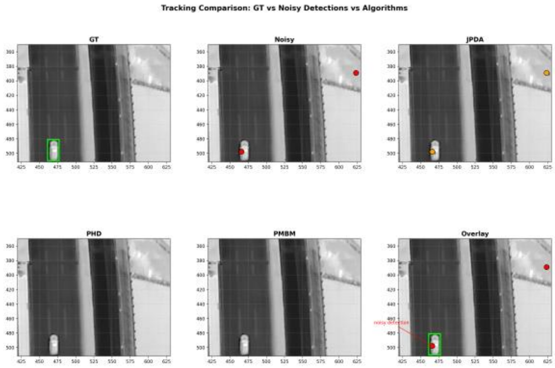
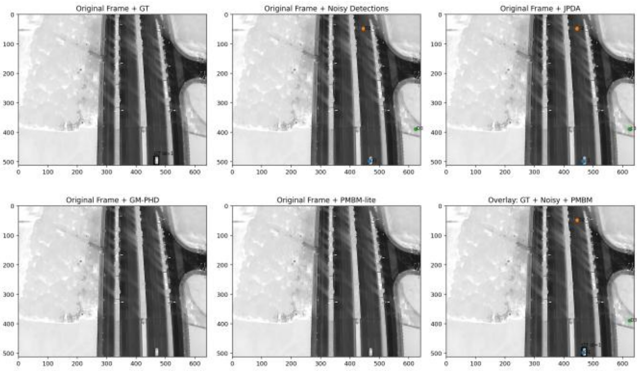
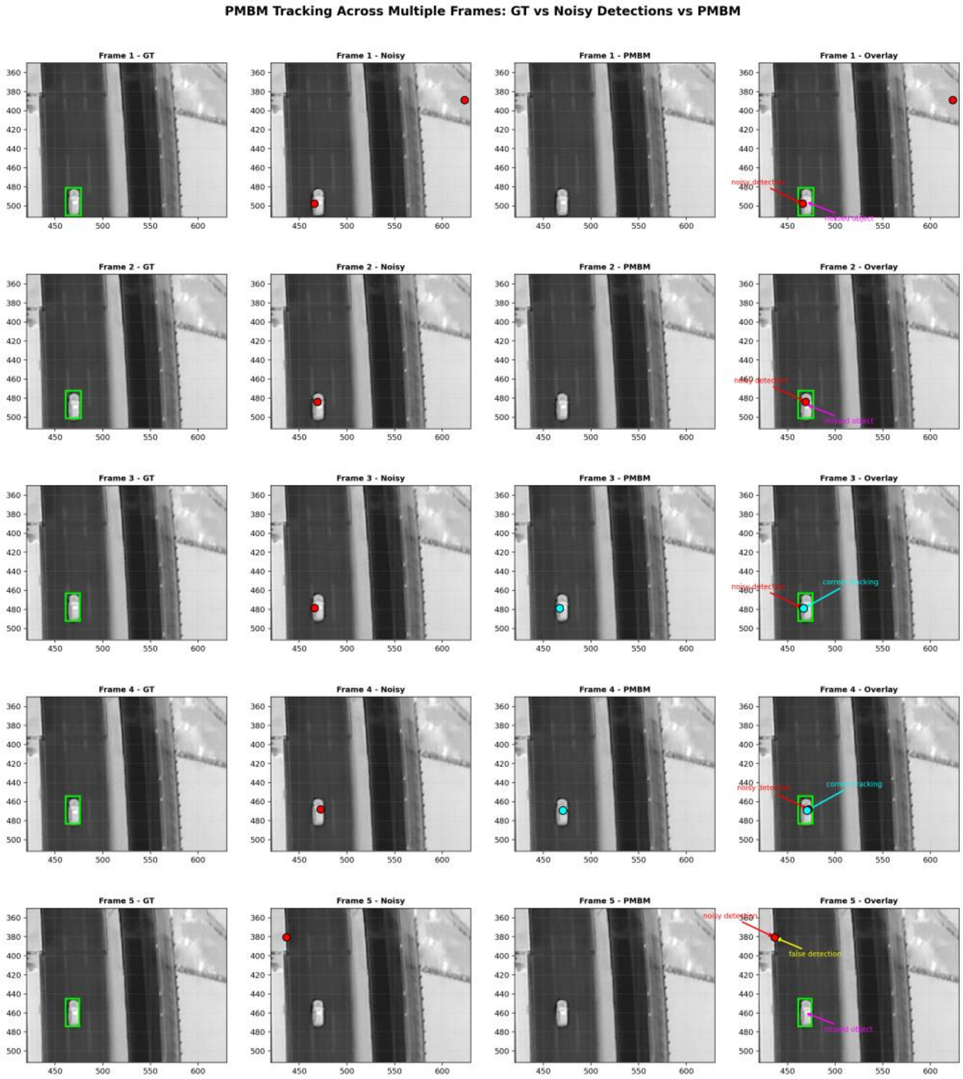
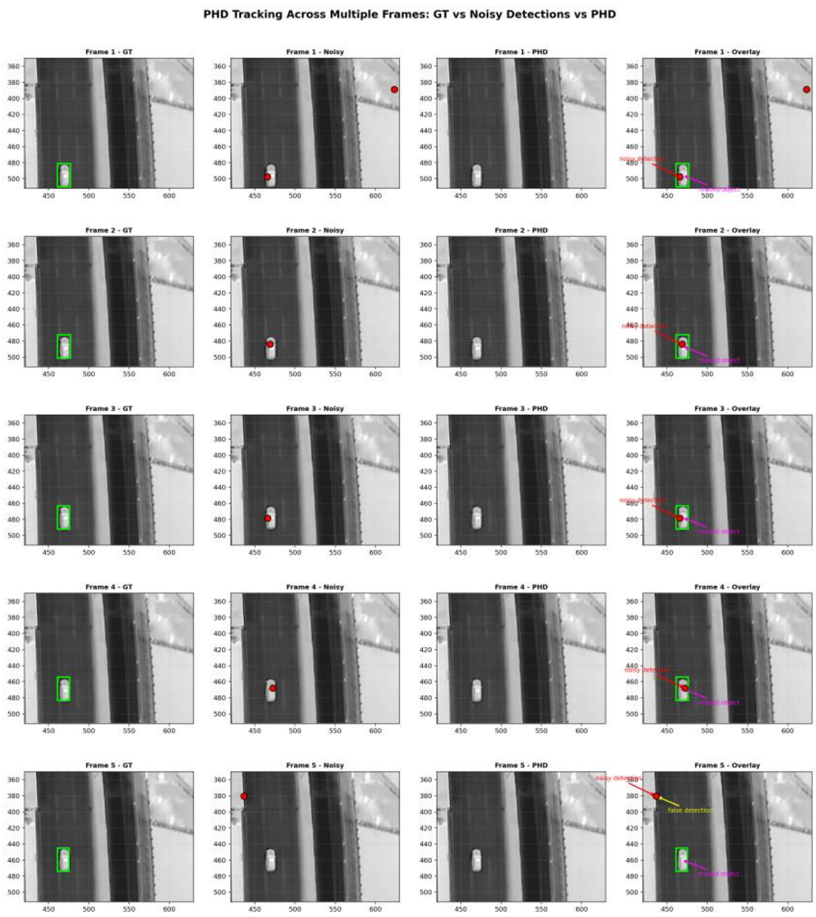
5. Visualization Analysis

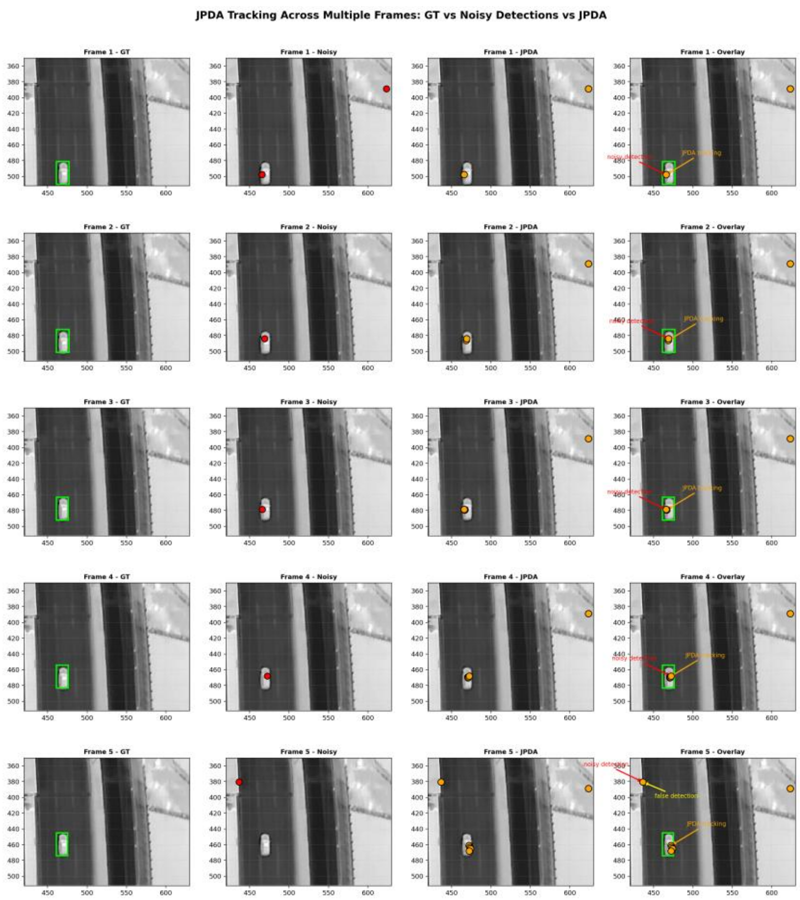
Visualizations were generated to analyze algorithm behavior across multiple frames. Figures show the comparison of ground truth, noisy detections, and tracking outputs for each algorithm.








PHD Tracking Across Multiple Frames: GT vs Noisy Detections vs PHD


JPDA Tracking Across Multiple Frames: GT vs Noisy Detections vs JPDA

The results show that:

- JPDA produces many false detections
- GM-PHD often loses objects
- PMBM maintains more stable tracking and recovers objects after temporary loss

Multi-frame visualization demonstrates that PMBM can miss objects in individual frames but successfully restores tracks later.

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6. Discussion

The results confirm that different algorithms have different strengths:

- JPDA prioritizes detecting all objects but introduces noise
- GM-PHD minimizes false detections but misses many objects
- PMBM-lite balances both aspects

This balance explains why PMBM achieves the best overall performance.

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7. Conclusion

In conclusion, PMBM-lite demonstrates the best performance among the evaluated algorithms.
It achieves the highest F1-score and the lowest KL divergence, indicating both accurate detection and good distribution modeling.
This makes PMBM-lite a strong choice for multi-object tracking in realistic conditions with noisy detections.
This demonstrates the importance of balancing detection accuracy and robustness in multi-object tracking systems.

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